

Question & Answers -2019

INTER SCIENCE (+2)

CHEMISTRY: 2019

[Time:3 Hours 15 Minutes]

[Full Marks:70]

Instructions for the candidates:

- Candidates are required to give their answers in their own words as far as practicable.
- Figures in the right margin indicate full marks.
- 15 Minutes of extra time has been allotted for the candidates to read the questions carefully.
- This question paper is divided into section- **Section-A and Section – B.**
- In **Section –A**, there are **35 Objective type questions** which are compulsory, each carrying **1 mark**. Darken the circle with blue/black ball pen against the correct option OMR Answer Sheet provided to you. **Do you use Whitener/Liquid/Blade/Nail etc. on OMR Sheet; otherwise the result will be invalid.**
- IN **Section – B**, there are **18 Short answer type questions** (each carrying **2 marks**), out of which any 10 questions are to be answered. Apart from this, there are **6 Long Answer Type question** (Each Carrying 5 marks), out of which any 3 questions are be answered.
- Use of any electronic appliances is strictly prohibited.

SECTION – A (Objective Type Questions)

Question No. 1 to 35 have four options, out of which only one is correct. You have to mark, your selected option, on the OMR – Sheet.

35×1= 35

1. Oleum is composed of H_2SO_4 and which constituent?

- (a) SO_2 (b) SO_3 (c) H_2S (d) $HSO_4^{(-)}$

Ans: Option (b)

2. The hybridization of N in NH_3 is

- (a) sp^3 (b) sp^2 (c) sp (d) d^2sp^3

Ans: Option (a)

3. The name of the compound OH is



- (a) Benzene hydroxide (b) Phenol
(b) Phenyl (d) Benzyl Alcohol

Ans: Option (b)

4. Drug used to reduce fever is called

- (a) Antipyretic (b) Benzyl Alcohol
(c) Antacid (d) None

Ans: Option (a)

5. Which of the following gas cannot be collected over water

- (a) O_2 (b) H_2S (c) SO_2 (d) N_2

Ans: Option (b)

6. Number of (σ) Sigma bonds in CH_4 is

- (a) 4 (b) 3 (c) 2 (d) 1

Ans: Option (a)

7. Second period of periodic table contains elements

- (a) 2 (b) 8 (c) 18 (d) 32

Ans: Option (b)

8. A process of heating one of a metal in excess of air below its melting points is called

- (a) Roasting (b) Smelting (c) Calcination (d) All

Ans: Option (a)

9. White and Yellow phosphorous are

- (a) Allotropes (b) Isomers
(c) Isobars (d) Isotones

Ans: Option (a)

10. Buna – N and Buna – S are

- (a) Natural rubber (b) Synthetic rubber
(c) Latex (d) Polythene

Ans: Option (b)

11. Dettol is used as

- (a) Disinfectant (b) Antiseptic

(c) Antimalarial

(d) All

Ans: Option (b)

12. Nitrogen & Oxygen are

(a) Metals

(b) Non- metals

(c) Metalloids

(d) None

Ans: Option (b)

13. Alcohol is formed when one of the hydrogen of Alkane is replaced by

(a) Hydroxyl group

(b) Ethanoic acid

(c) Propanoic acid

(d) Methanol

Ans: Option (a)

14. IUPAC name of CH_3COOH is

(a) Methanoic acid

(b) Ethanoic acid

(c) Propanoic acid

(d) Methanol

Ans: Option (b)

15. Formula of Sucrose is

(a) $C_{12}H_{22}O_{12}$

(b) $C_{12}H_{22}O_{11}$

(c) $C_{12}H_{24}O_{11}$

(d) $C_{12}H_{22}O_{10}$

Ans: Option (b)

16. Which of the following gas is mono – atomic?

(a) Chlorine

(b) Helium

(c) Nitrogen

(d) Oxygen

Ans: Option (b)

17. $K_3[Fe(CN)_6]$ is a

(a) Double salt

(b) Complex compound

(c) Acidic Salt

(d) Common salt

Ans: Option (b)

18. Transition elements are also known as

(a) *s* – block elements

(b) *p* – block elements

(c) *d* – block elements

(d) *f* – block elements

Ans: Option (c)

19. Ammonia changes the moist and red litmus paper into

- (a) Blue (b) Green (c) Black (d) white

Ans: Option (a)

20. Nitric acid is prepared by

- (a) Contact process (b) Ostwald's
(c) photosynthesis (d) Haber's process

Ans: Option (b)

21. PCl_5 is a

- (a) Oxidising agent (b) Reducing agent
(c) Oxidizing and Reducing agent both
(d) None

Ans: Option (b)

22. The oxidation number of Fe in $K_4[Fe(CN)_6]$ is

- (a) +2 (b) +3 (c) -2 (d) -3

Ans: Option (a)

23. Chief ore of iron is

- (a) Magnetic (b) Siderite
(c) Haematite (d) All

Ans: Option (c)

24. H_2SO_4 is a

- (a) Dibasic acid (b) Monobasic acid.
(c) Mono acid base (d) Di acid base

Ans: Option (a)

25. Over production of acid in stomach can be controlled by

- (a) Antacid (b) Ethene (c) Propene (d) Ethyne

Ans: Option (a)

26. Polythene is a polymer of

- (a) Ethane (b) Ethene (c) Propene (d) Ethyne

Ans: Option (b)

27. The symbol of Helium is

- (a) He (b) Hi (c) Hm (d) All

Ans: Option (a)

28. Carbohydrate which yields three to ten mono saccharide units on hydrolysis is called

- (a) Monosaccharide (b) Disaccharide
(c) Oligosaccharide (d) Polysaccharide

Ans: Option (c)

29. Group I elements of Periodic Table are known as

- (a) Alkali metals (b) Alkaline earth metals
(c) Inert gases (d) None

Ans: Option (a)

30. pH of pure water is

- (a) 1 (b) 4 (c) 3 (d) 7

Ans: Option (d)

31. $\text{CH}_3\text{CH}_2\text{NH}_2$ is known as

- (a) Ethyl amine (b) Propyl amine
(c) Chromatography (d) None

Ans: Option (a)

32. one Faraday electricity is equal to how much of Coulombs?

- (a) 96550 (b) 96500 (c) 96000 (d) 95500

Ans: Option (b)

33. Copper metal is purified by which of the following method.

- (a) Electrolytic refining (b) Zone refining
(c) Chromatography (d) None

Ans: Option (a)

34. The formula of Nitric acid is

- (a) HNO_3 (b) $\text{H}_2\text{N}_2\text{O}_2$ (c) HNO_2 (d) HNO_4

Ans: Option (a)

35. The functional group of alcohol is

- (a) $-\text{COOH}$ (b) $-\text{CHO}$
(c) $-\text{C}=\text{O}$ (d) $-\text{OH}$

Ans: Option (D)

SECTION – B (Non – objective Type Questions)

Short Answer Type Questions

Question No. 1 to 18 are short answer type. Answer any 10 question. Each Question carries 2 marks.

$(10 \times 2 = 20)$

1. What do you mean by point defect in a crystal?

Ans: Point defect – The defects naming –(a) Schottky defect, (b) Non- Schottky defect, and the impurities are combinedly known as the point defect.

Frenkel defect – This type of defect which is created when an ion leaves its correct site and occupies an interstitial state.

It does not give any change in density

This type of defect has been found in silver halides (AgCl , AgBr & AgI).

2. What do you mean by Primary cell?

Ans: Such cells in which redox reaction occurs once, it is not rechargeable. It means in primary cells the electrode reaction cannot be reversed by an external electrical energy source and it becomes dead over a period of time, example-dry cell, mercury cell etc.

3. Write first law of Faraday's law of electrolysis.

Ans: Faraday's law of electrolysis-mass of substance liberated or deposited at electrodes is directly proportional to the amount of electricity passed through the electrolyte, during electrolysis

$$m \propto q$$

$$[I = \frac{q}{t}]$$

$$m = z \times I \times t \quad [z = \text{electrochemical euivalence}]$$

4. Explain order of Reaction.

Ans: The order of a chemical reaction with respect to each reactant is defined as the exponent to which the concentration term of that reactant. In the rate law is raised.

$$\text{Rate} = k[A]^x \cdot [B]^y$$

Order for conc. Of A = x

Order for con. Of B = y

5. Conc. HNO_3 is stored in Aluminium vessel, why?

Ans: Conc. HNO_3 does not react with aluminium hence Conc. HNO_3 is stored in aluminium vessel.

6. Sulphur forms SF_6 but not SCl_6 why?

Ans: SF_6 is thermally stable and chemically inert gas. Thus sulphur forms SF_6 but not SCl_6

7. Find the oxidation number of Cu in the complex ion: $[Cu(NH_3)_4]^{2+}$

Ans: Suppose O.N of Cu = x

$$\text{O.N of } NH_3 = 0$$

$$\text{Then, } [Cu(NH_3)_4]^{2+} = 2$$

$$x + 4 \times 0 = +2$$

$$x = +2$$

Therefore, O.N of Cu in $[Cu(NH_3)_4]^{2+}$ is +2

8. Name of element present in Hemoglobin.

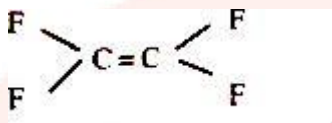
Ans: Hemoglobin contain iron and it is also like protein and protein are made of amino acids. All amino acids contain the elements CHNO.

9. what do you mean by BOD and COD?

Ans: The amount of oxygen required by bacteria to break down the organic matter present in a certain volume of a sample of water is called Bio-chemical Oxygen Demand (BOD). The amount of oxygen consumed by the pollutants is called COD of the given sample of water.

10. Write the monomere of Teflon?

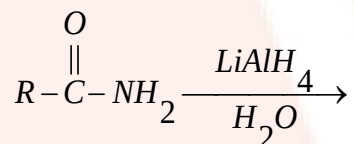
Ans: Tetra fluoro ethane



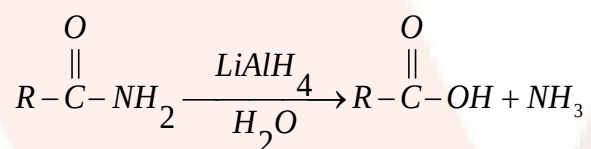
11. What are Amino acids?

Ans: Amino acids are organic component containing $(-NH_2)$ and carboxylic $(-COOH)$ group adverse nature. Glycine, Alanine etc. Tyrosine was 1st obtained from cheese.

12. Complete the reaction



Ans:



13. Alkyl amine is stronger base than Ammonia, why?

Ans: Due to e^- Relasing to inductive effect (+I effect) of alkyl grps, it (R) pishes e^- towards N and thus makes the unshared e^- pair more available for sharing wih proton of acid.

Makeover the subtituted ammonium formed from amine get stabished due to dispersal of +ve charge by + I effect of alkyl .

Hence $R-NH_2$ more base than NH_3 .

14. Define Osmotic pressure.

Ans: Osmotic Pressur: The excess of pressure on the side of soluition. That stops the net of solvent from lower concentration to higher concentration solution via semipressure membrance.

15. What are Enzymes?

Ans: Enzymes are biological catalysts. Almost all enzymes are globular proteins. They are very specific for a particular reaction and for a particular substrate. They are generally named after the compounds. Maltose.

16. Silver is a Good conductor of electricity . Why?

Ans: Silver has the lowest contact resistance. It has the highest thermal conductivity of any element and the highest light reflection. Silver is good conductor of electricity and heat.

17. What are ideal and non – ideal solutions?

Ans: Ideal solution : Those solutions which obey Raoult's law at all concentrations are called ideal solutions.

$$\begin{aligned} \text{For ideal solution } \Delta H_{\text{mixing}} &= 0 \\ \Delta V_{\text{mixing}} &= 0 \end{aligned}$$

n – Hexane + n – Heptane.

Non-ideal solution: Those solutions which do not obey Raoult's law at all concentration are called non-ideal solution.

$$\begin{aligned} \text{For non ideal solution } \Delta H_{\text{mixing}} &= 0 \\ \Delta V_{\text{mixing}} &= 0 \end{aligned}$$

Acetone chloroform.

18. What are the advantages of using Cryolite in the electrolytic reduction of Alumina?

Ans: In the metallurgy of alumina, the metal to be isolated from alumina (Al_2O_3) by carrying out its electrolytic reduction. The melting point of alumina as such is 2323 K. It is therefore mixed with Cryolite (Na_3AlF_6) which lowers its melting point to 1173 K. Moreover, Cryolite also increases the electrical conductivity of alumina.

LONG ANSWER TYPE QUESTIONS

Question No. 19 to 24 are long Answer Type Question. Each question carries 5 marks.
Answer any 3 question. 3×5=15

19. Define Osmotic pressure. Prove that Osmotic pressure is a colligative property.

Ans: Osmotic pressure- The excess pressure that must be applied to the solution side to prevent the passage of solvent into it through a semi-permeable membrane.

According to Van't Hoff-

Osmotic pressure, $\pi \propto C$ (at certain temperature).....(i)

And Osmotic pressure, $\pi \propto T$ (ii)

$$\Rightarrow \pi \propto CT \quad \Rightarrow \pi \propto \frac{n}{V}T \quad \Rightarrow \pi = \frac{n}{V}RT$$

Where, n= Number of mole, R= Gas constant

Hence, Osmotic pressure depends upon modal concentration not on solute nature. Hence, Osmotic pressure is molecular property.

20. Find the expression for first order reaction.

Ans: . Let us consider $R \rightarrow \text{product}$

$$t = 0 \rightarrow a \rightarrow 0$$

$$t = t \rightarrow (a-x) \rightarrow x$$

$$\frac{dx}{dt} \propto (a-x) \Rightarrow \frac{dx}{dt} = k(a-x)$$

$$\text{integrating } \int \frac{dx}{(a-x)} = \int k dt$$

$$\Rightarrow -\lim(a-x) = kt + \frac{c}{v} \dots \dots (i)$$

$$\text{when } t=0, x=0 \Rightarrow c = -\lim a$$

$$-\lim(a-x) = kt - \lim 0$$

$$kt = \lim a - \lim(a-x) = \lim \frac{a}{(a-x)}$$

$$k = \frac{1}{t} \lim \frac{a}{(a-x)} \rightarrow k_s = \frac{2.303}{t} \log \frac{a}{(a-x)}$$

$$\text{unit} = \frac{\tan \alpha}{\sec \alpha \cot \alpha} = \tan^{-1} c = \sec^{-1}$$

Half life:

$$x = \frac{a}{2} \quad \text{when } t=t_{1/2}$$

$$k = \frac{2.303}{t_{1/2}} \log \frac{a}{\left(a - \frac{a}{2}\right)} = \frac{2.303}{t_{1/2}} \log 2$$

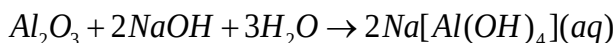
$$\Rightarrow t_{1/2} = \frac{2.303}{k_s} \log 2 = \frac{2.303}{k_I} \times 0.3010$$

$$\Rightarrow t_{1/2} = \frac{0.693}{k}$$

21. Write in brief how Aluminium is extracted from Bauxite?

Ans: Important ores of aluminium are bauxite ($Al_2O_3 \cdot 2H_2O$) and cryollite Na_3AlF_6

Principle of extraction of aluminium from bauxite –

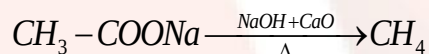


22. What happens when:

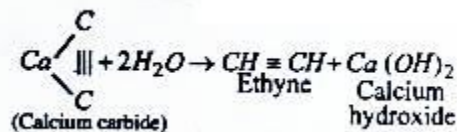
(a) Sodium acetate is heated with Soda lime?

(b) Calcium carbide is allowed to react with water?

Ans: (a) when sodium acetate is heated with soda- lime, Methane gas is liberated.



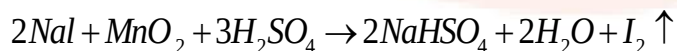
(b) When Calcium carbide reacts with water, it liberates ethyne gas



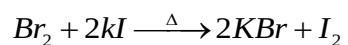
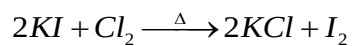
23. What are the main sources of Iodine? How it is extracted from sea weeds?

Ans: Nature sources of iodine: Due to its reactivity iodine is not found in nature in free state. Its main sources are (i) Sea weeds (ii) Chile salt lakes (iii) Natural brine.

Extraction of Iodine from sea weeds: Sea weed, laminaria contains iodine. Sea weed is well dried and burnt in deep pits carefully so that iodine is not destroyed. The obtained ash is called 'kelp'. Which contains 0.4 to 1.3% iodine. Kelp is dissolved in water and solution is partially crystallised when less soluble KI and NaI remain in the mother liquor. Conc. H_2SO_4 is added when basic sulphides deposit at the bottom, which is filtered and removed. Now the filtrate is mixed with MnO_2 and Conc. H_2SO_4 and heated in an iron vessel. Iodine vapourises due to the reaction and is collected in Aludel. Iodine is now collected as solid after condensation.



Iodine obtained by this method contains Cl_2 and Br_2 as impurities. It is treated with KI to obtain pure iodine.

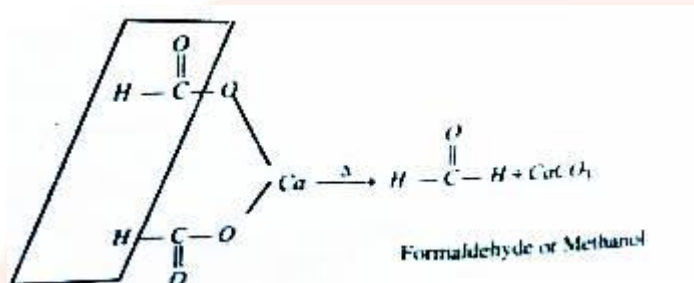


24. What happens when:

(a) Calcium formate is heated strongly?

(b) Methanol reacts with Ammonia?

(a) When Calcium formate heated strongly to give formaldehyde.



(b) When Methanol reacts Ammonia to give Methanamine.

